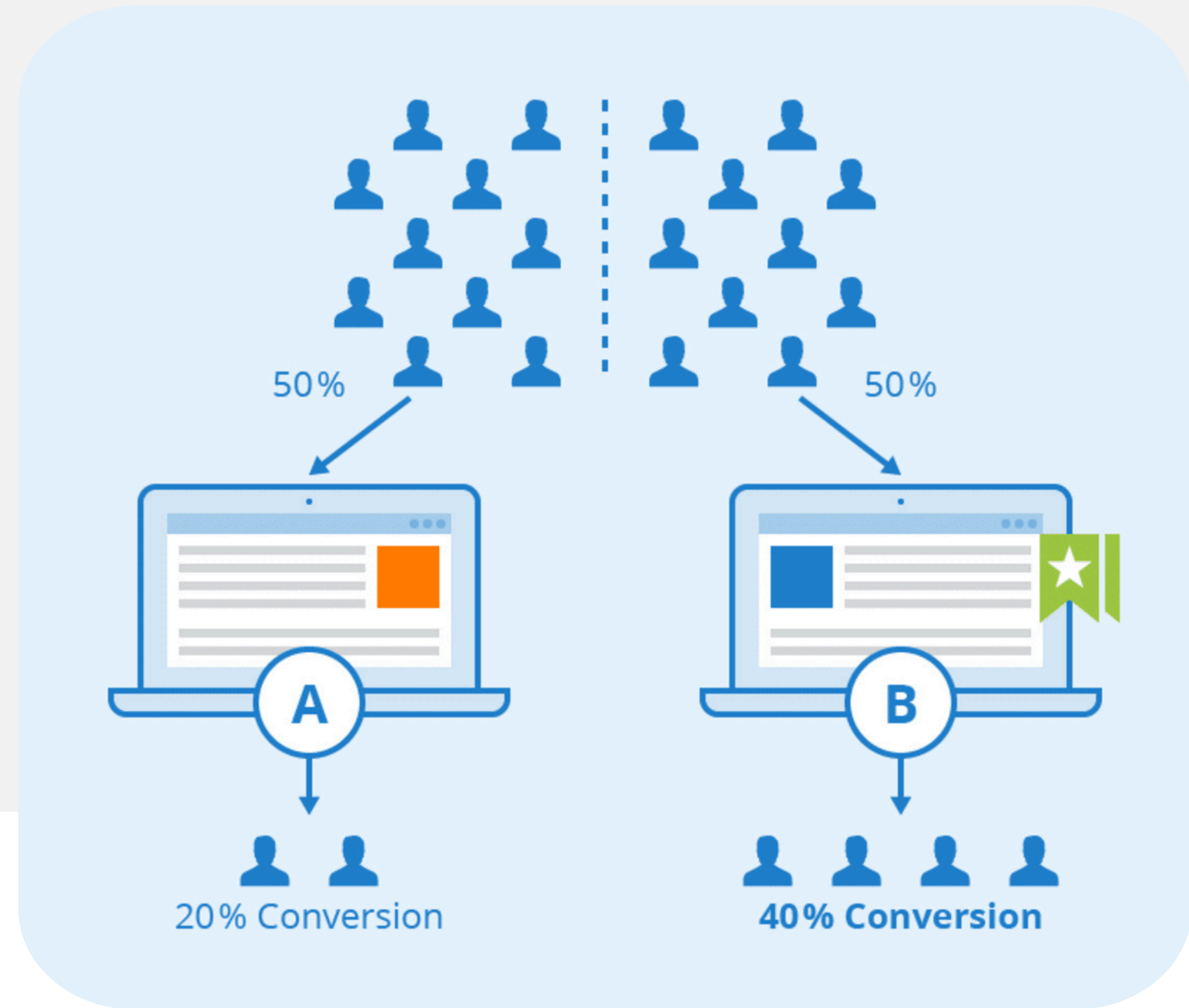


A/B Testing Analysis of Website Page Performance

Through Page Type



Agenda

- Problem Statement
- Research Question
- Data Overview
- Methodology
- Analysis and Findings
- Hypothesis Testing
- Recommendations

Problem Statement

In the competitive digital landscape, improving website conversion rates is essential for business success.

This study uses data-driven analysis to evaluate whether the new webpage design leads to higher user conversions compared to the old page, helping determine which version performs better.



Research Question

Is there a relationship between page type and user conversion behavior?

Can the new page design improve user engagement and conversions without negatively affecting the overall user experience?

Data Overview

For this analysis, we used an A/B testing dataset containing user interaction records. Data cleaning was performed to keep only the relevant columns needed for the analysis. These columns include user ID, treatment group, page version, time spent on the page, and conversion outcome.

Relevant columns used for this research:

- **id:** Unique identifier.
- **con_treat:** control group or treatment group.
- **page:** (old_page or new_page).
- **minute:** Time spent by the user on the page (in minutes).
- **converted:** Indicates whether the user completed the conversion (1 = yes, 0 = no).

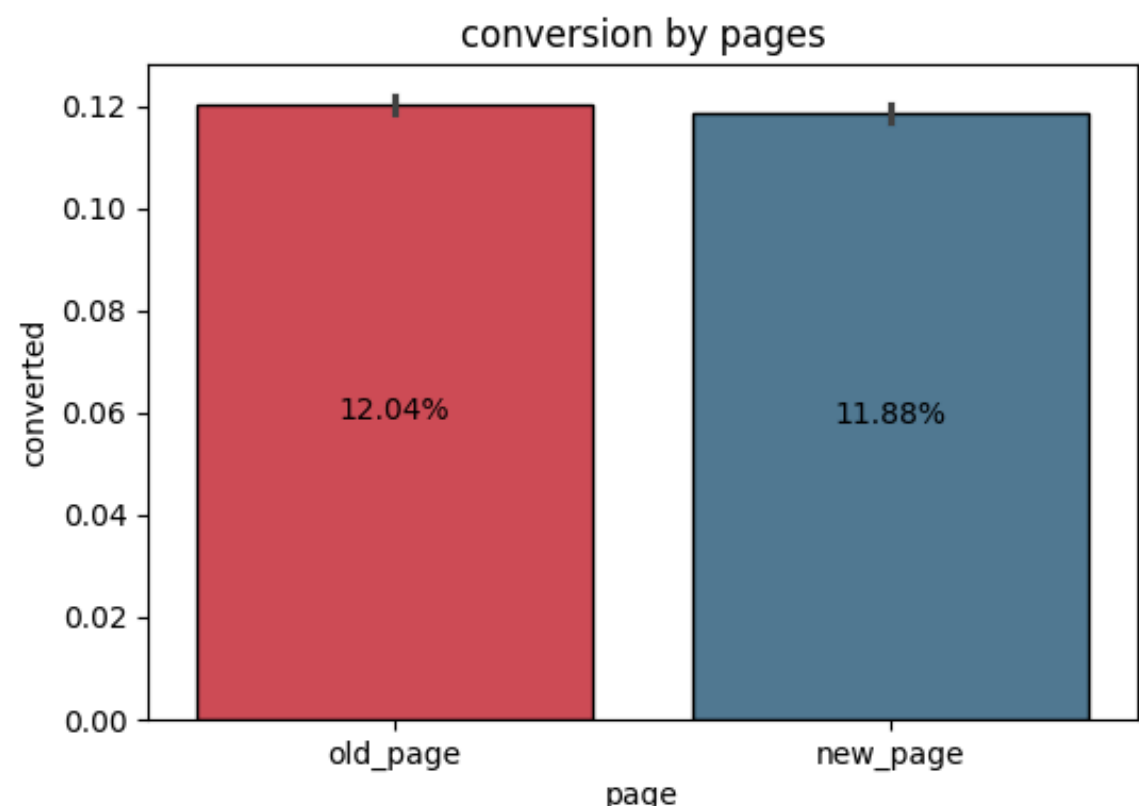
	id	con_treat	page	minute	converted
0	851104	control	old_page	11	0
1	804228	control	old_page	1	0
2	661590	treatment	new_page	55	0
3	853541	treatment	new_page	28	0
4	864975	control	old_page	52	1

Methodology

Step	Description
Descriptive Analysis	Performed statistical analysis to summarize key aspects of the data, focusing on time spend , converted and page version
Hypothesis Testing	Conducted a T-test to evaluate whether there is a significant difference in conversion rates between the control group (old page) and the treatment group (new page). The hypothesis tested whether the new page design leads to a higher conversion rate compared to the old page.

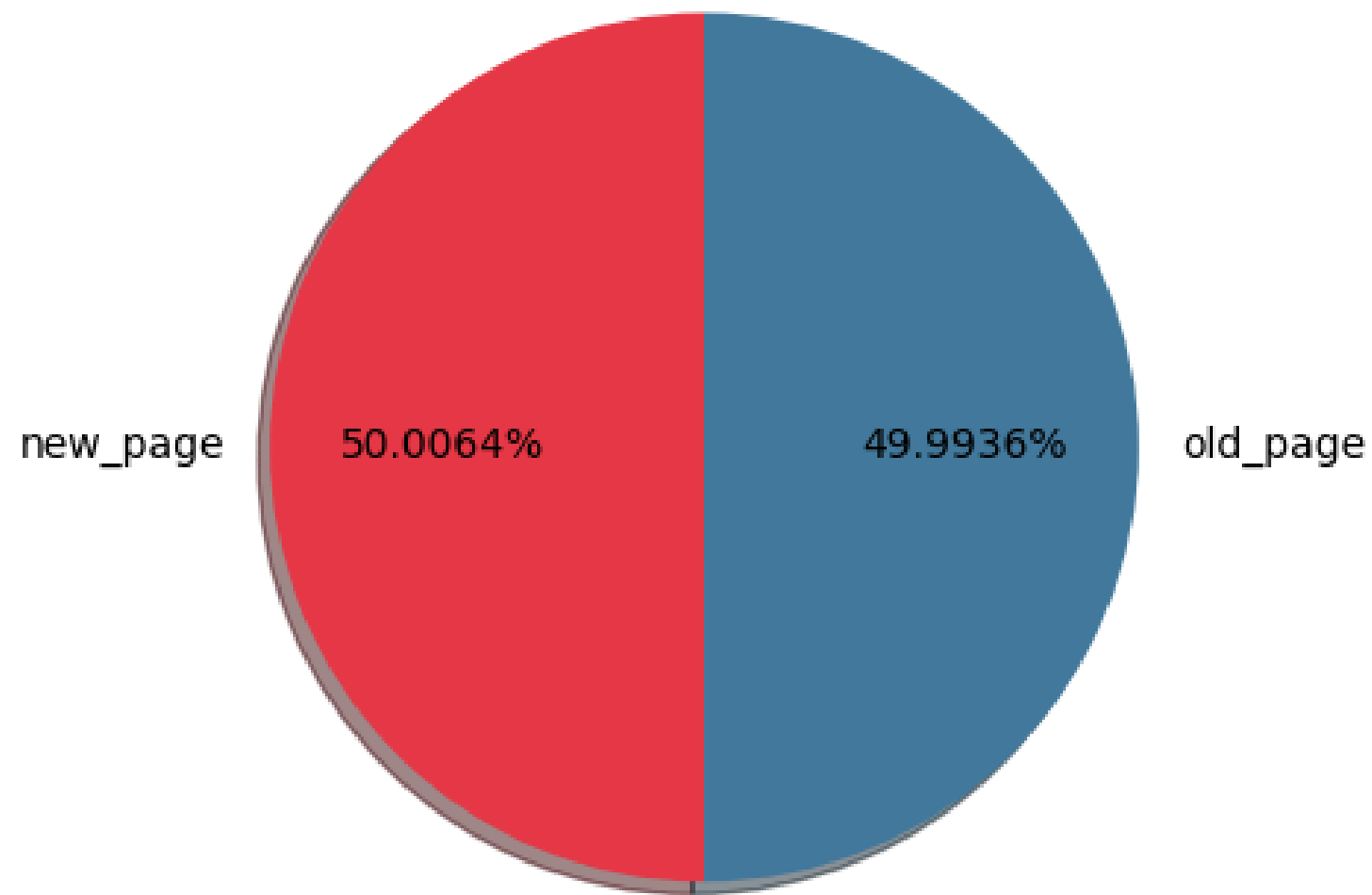
Journey Insights

- **Marginal Difference:** The old_page achieved a conversion rate of 12.04%, while the new_page achieved 11.88%.
- **Performance Lead:** The old page slightly outperformed the new page by a narrow margin of 0.16%.
- **Statistical Significance:** The black error bars (confidence intervals) at the top of each bar are overlapping. This suggests that the difference between the two pages is likely not statistically significant.



	page version	Mean	Standard Deviation
converted	old_page	0.12	0.33
	new_page	0.12	0.32
time spent	old_page	29.58	17.28
	new_page	29.53	17.32

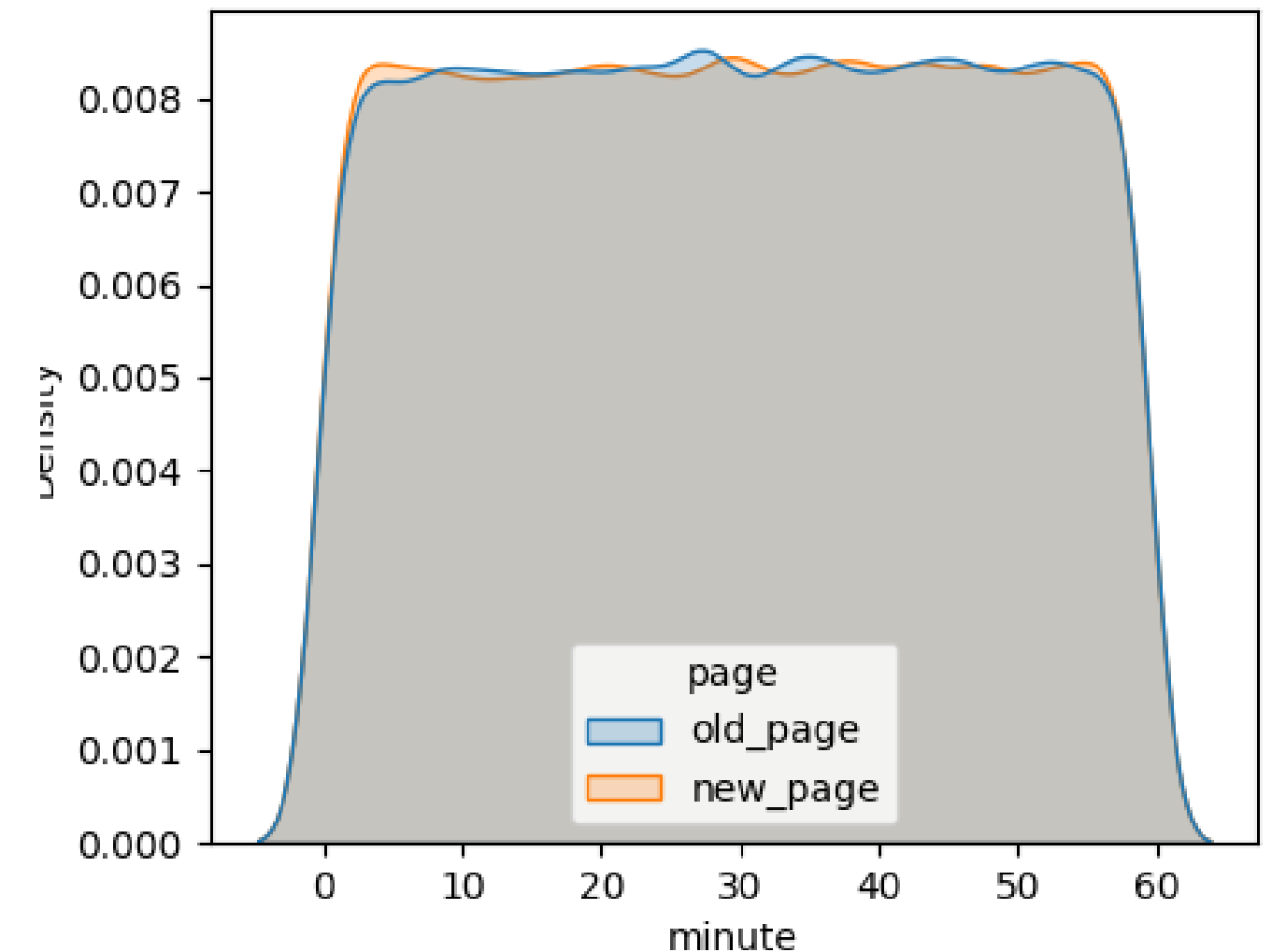
Page Distribution



- The pie chart shows the distribution of users between the two page versions in the A/B test. The new_page accounts for 50.0064% of users, while the old_page represents 49.9936%.
- This nearly equal distribution indicates that users were evenly split between the control and treatment groups, ensuring a balanced experiment for accurately comparing conversion outcomes.

Passenger Count Analysis

- The KDE plot shows the distribution of time (in minutes) users spent on the old_page and new_page.
- The curves for both pages almost completely overlap, indicating very similar time distributions.
- This suggests that the page version does not significantly affect how long users stay on the page.



Hypothesis Testing

Null hypothesis: There is no difference in conversion rates between the old page and the new page.

Alternative hypothesis: There is a difference in conversion rates between the old page and the new page.

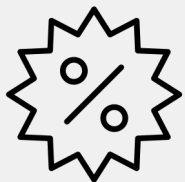
With a T-statistic of 0.87 and a P-value of 0.384 (> 0.05), we fail to reject the null hypothesis, suggesting that there is no statistically significant difference in conversion rates between the old page and the new page.

Recommendations



Retain the Current Page Design

Since the P-value (0.384) is greater than 0.05, there is no statistically significant difference between the old and new page. Therefore, it is recommended to continue using the current (old) page until a design proves to significantly improve conversions.



Increase User Engagement and Time Spent

Add interactive elements, clearer call-to-action buttons, and better content to encourage users to stay longer on the page, which may increase the chances of conversion.



Provide a Secure and Seamless Experience

Ensure the page loads quickly and offers a secure, smooth browsing experience to build user trust and improve overall usability.

Thank you!
